

Here are the formal notes for the requested video.

Subject: General Linear Models (Part 2): Using Linear Models for t-tests and ANOVA

Source: StatQuest with Josh Starmer http://www.youtube.com/watch?v=NF5_btOaCig

I. Introduction & Objective

This video builds upon the concepts of Linear Regression (from Part 1) to demonstrate how the **General Linear Model (GLM)** framework can be used to perform **t-tests** and **Analysis of Variance (ANOVA)** [00:31].

- **Unifying Principle:** The core idea is that the same fundamental technique (minimizing sums of squares, calculating an F-statistic) can be applied to both regression (continuous predictors) and group comparisons (categorical predictors).
- **Key Tool:** This unification is achieved by using a **Design Matrix** [00:38].

II. The General Linear Model Framework

The process for calculating a p-value in a GLM, whether for regression, a t-test, or ANOVA, follows the same sequence of steps [01:53]:

1. **Calculate SS_{mean} (Total Variation):** Fit a "null model," which is the **overall mean** of all data points, ignoring all predictors or groups. Calculate the sum of squared residuals around this overall mean [02:24].
2. **Calculate SS_{fit} (Unexplained Variation):** Fit the "full model" to the data.
 - For **Regression**, this is the least-squares regression line.
 - For **t-tests/ANOVA**, this involves fitting the mean for *each category* [03:13, 03:40]. Calculate the sum of squared residuals around this full model.
3. **Calculate the F-statistic:** Use the identical formula from linear regression to compare the variance explained by the full model versus the null model.

$$F = \frac{(SS_{\text{mean}} - SS_{\text{fit}})/(p_{\text{fit}} - p_{\text{mean}})}{SS_{\text{fit}}/(n - p_{\text{fit}})}$$

4. **Find the p-value:** Convert the F-statistic to a p-value using the F-distribution.

III. Application 1: The t-test as a Linear Model

A t-test compares the means of two groups to see if they are significantly different [01:32].

Example: Comparing gene expression between *Control* mice and *Mutant* mice.

1. Calculate SS_{mean} :

- The "null model" is the overall mean expression of *all* mice (both control and mutant) combined [02:16].
- The number of parameters in this model is $p_{\text{mean}} = 1$ (the overall mean) [08:03].

2. Calculate SS_{fit} :

- The "full model" fits a separate mean for each group.
- The least-squares fit for the *Control* data is the **mean of the Control group** [03:13].
- The least-squares fit for the *Mutant* data is the **mean of the Mutant group** [03:40].
- SS_{fit} is the sum of squared residuals calculated around each group's *own mean* [07:25].
- The number of parameters in this model is $p_{\text{fit}} = 2$ (one mean for controls, one mean for mutants) [08:28].

3. Calculate F-statistic:

- Numerator d.f. = $p_{\text{fit}} - p_{\text{mean}} = 2 - 1 = 1$.
- Denominator d.f. = $n - p_{\text{fit}} = n - 2$.
- Plugging these values into the F-statistic formula yields the p-value for the t-test [08:36].

The Design Matrix

To represent this two-mean model as a *single equation* for a computer to solve, a **design matrix** is used [04:13]. This matrix uses 1s and 0s as "on/off switches" to specify which parameter applies to which data point [06:05].

• Abstract Equation:

$$y_i = (\text{Mean}_{\text{control}}) \times (\text{Col}_1) + (\text{Mean}_{\text{mutant}}) \times (\text{Col}_2) + \text{residual}_i$$

• Design Matrix:

- For a *Control* mouse: $[1 \ 0]$ (activates $\text{Mean}_{\text{control}}$) [04:51].
- For a *Mutant* mouse: $[0 \ 1]$ (activates $\text{Mean}_{\text{mutant}}$) [05:47].

This flexible matrix-based approach allows the same software to solve regression, t-tests, and more complex models.

IV. Application 2: ANOVA as a Linear Model

ANOVA tests if the means of *three or more* categories are the same [09:22].

Example: Comparing gene expression across 5 categories (e.g., Control, Mutant, Control+Diet, Mutant+Diet, Heterozygote) [09:31].

The process is identical to the t-test, but scaled:

1. Calculate SS_{mean} :

- The "null model" is the overall mean expression of *all* mice from *all* 5 groups combined [09:45].
- $p_{\text{mean}} = 1$ (the overall mean) [10:01].

2. Calculate SS_{fit} :

- The "full model" fits a separate mean for *each of the 5 categories* [10:09].
- SS_{fit} is the sum of squared residuals calculated around each category's *own mean*.
- The number of parameters is $p_{\text{fit}} = 5$ (one mean for each category) [10:20].

3. The Design Matrix:

- The design matrix for this model would have **5 columns**, one for each category, to act as on/off switches [10:20].

4. Calculate F-statistic:

- Numerator d.f. = $p_{\text{fit}} - p_{\text{mean}} = 5 - 1 = 4$.
- Denominator d.f. = $n - p_{\text{fit}} = n - 5$.
- This calculation yields the F-statistic and the p-value for the ANOVA [10:36].

V. Final Note on Design Matrices

The design matrices shown in the video (e.g., $\begin{bmatrix} 1 & 0 \end{bmatrix}$ and $\begin{bmatrix} 0 & 1 \end{bmatrix}$) are conceptually simple but are *not* the standard matrices used in most statistical software [10:53]. Software typically uses a different but mathematically equivalent formulation (e.g., one column for an intercept and other columns for the *differences* from that intercept). Both methods achieve the same result [11:08].

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